Temporal Fractal Coupling of Volatility and Trading Volume: A Robust Pattern Without Forecast Power in S&P 500 Stocks, 2015–2026

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Abstract

Do the fractal scaling properties of price volatility and trading volume co-evolve, and does their co-movement carry predictive information about future market conditions? I apply detrended fluctuation analysis to 50 S&P 500 constituents over 2015–2026, computing rolling Hurst exponents for absolute returns ($H^{(i)}_{|r|,t}$) and log-volume ($H^{(i)}_{v,t}$) in aligned windows, and introduce the Coupling Intensity Index (CII), a rolling Pearson correlation that measures their temporal co-movement. The cross-sectional relationship between full-sample exponents is null ($r = -0.02$), but temporal coupling within firms is strong and positive in 49 of 50 equities (mean $r = 0.665$). The coupling intensifies during high-VIX regimes ($p = 0.036$) and nearly doubles during the COVID-19 crisis. The predictive question, however, is more nuanced than the descriptive one. Under inference robust to two-way clustering (Cameron et al., 2011), Bell–McCaffrey CR2 with Satterthwaite degrees of freedom (Imbens and Kolesár, 2016), and the wild cluster restricted bootstrap (MacKinnon and Webb, 2017), CII has no incremental predictive content for forward Amihud (Amihud, 2002) illiquidity computed in the standard dollar-volume convention, nor for realised volatility, the Corwin–Schultz spread (Corwin and Schultz, 2012), the Roll spread (Roll, 1984), vol-of-vol, max drawdown, or abnormal turnover. An apparently strong predictive result against a non-standard share-volume illiquidity proxy dissipates

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once both proper Amihud convention and dependence-robust inference are imposed; I document and dissect this as a price-level confound. The contribution is therefore a robust empirical pattern of within-firm fractal coupling, regime-amplified during stress, without demonstrated forecasting power for standard market outcomes, alongside a methodological caveat for the wider econophysics-meets-finance literature: distinguish share-volume from dollar-volume liquidity proxies, and pair small samples with cluster-robust small-sample inference.

Keywords: Hurst exponent, Detrended fluctuation analysis, Price-volume relationship, Long memory, Coupling Intensity Index, Amihud illiquidity, Cluster-robust inference, Wild cluster bootstrap, Methodological caveat

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1. Introduction

Financial price volatility and trading volume each exhibit long-range dependence.[2] Absolute returns display Hurst exponents of approximately 0.7–0.8 (Ding et al., 1993; Bollerslev and Jubinski, 1999), while volume persists even more strongly at $H \approx 0.7$–0.9 (Lobato and Velasco, 2000). Whether these two fractal regimes are *coupled* (whether they co-evolve over time, and whether that co-evolution carries information a financial economist would care about) has not been systematically investigated.

This paper addresses both questions. I distinguish between *static* coupling (the cross-sectional correlation of Hurst exponents across firms) and *temporal* coupling (the within-firm co-movement of rolling Hurst exponents over time), and I construct a Coupling Intensity Index (CII) that measures the strength of temporal coupling as a time-varying quantity. I then test four hypotheses:

H1 *Static coupling:* Firms with more persistent volume also have more persistent volatility in the cross-section.

[2]This paper builds on an earlier MSc project (Malempati Hari, 2013) in which I examined whether Hurst-based fractal characteristics of stock-price series were associated with average trading volume in the cross section. That earlier work found a weak but statistically significant static relationship. The present study extends that line of inquiry by shifting from static association to temporal co-evolution, asking whether the persistence structures of volatility and volume move together through time and whether that coupling predicts future market illiquidity.

H2 *Temporal coupling:* Rolling Hurst exponents for $|r_t|$ and $\log V_t$ co-move positively within firms over time.

H3 *Regime dependence:* Temporal coupling intensifies during high-uncertainty regimes (elevated VIX).

H4 *Predictive content:* CII has incremental predictive content for standard forward liquidity and volatility measures.

The results are as follows. H1 is rejected: the cross-sectional correlation is $r = -0.02$ ($p = 0.88$), meaning that a firm's volatility persistence does not reveal a meaningful relationship with its volume persistence in this sample. H2 is strongly supported: mean temporal coupling is $r = 0.665$, with 49 of 50 equities exhibiting significant positive co-movement. H3 is confirmed: coupling is significantly higher in high-VIX regimes (Mann-Whitney $p = 0.036$) and nearly doubles during the COVID-19 crisis.

H4, by contrast, is rejected. Under inference robust to within-firm and within-time-period dependence, CII has no incremental predictive content for forward Amihud illiquidity computed in the standard dollar-volume convention of Amihud (2002), nor for realised volatility, the Corwin–Schultz spread, the Roll spread, the volatility of realised volatility, the maximum drawdown, or abnormal turnover. A specification that uses share volume rather than dollar volume in the Amihud denominator does produce a statistically significant CII coefficient ($t_{\text{two-way}} = 2.99$, $p = 0.004$) and survives the wild cluster restricted bootstrap ($p = 0.018$); but this signal disappears once dollar volume replaces share volume in the denominator, as standard practice requires. Section 5.5.4 dissects the divergence as a price-level confound: the share-volume Amihud co-moves with within-firm price-level drift, and CII correlates with that drift component rather than with price impact per dollar traded. Once the confound is removed, the predictive null is robust across ten target specifications and three modern small-sample inference methods.

This negative result is not a stylistic disappointment; it sharpens the paper's contribution. Prior work on price-volume fractal relationships (Podobnik et al., 2009) and common long-memory components (Bollerslev and Jubinski, 1999) has established that volatility and volume share structural features. The present paper shows that the within-firm coupling of those structural features (a) exists in a strong and quantitatively measurable form, (b) is regime-dependent and amplifies during stress, and (c) does not, on this sample and at this frequency, translate into demonstrable predictive content

for standard liquidity or volatility outcomes. The empirical claim is therefore narrower than the original H4 hypothesis but is sharper, harder to attack, and methodologically more honest. The paper also delivers a transferable cautionary lesson: in the moderately-clustered settings typical of empirical asset pricing, the choice between share-volume and dollar-volume illiquidity proxies and the use of dependence-robust small-sample inference together determine whether one finds a signal or a confound.

The conceptual distinction that organises the paper is between first-order and second-order fractal properties. Detrended cross-correlation analysis (DCCA; Podobnik and Stanley, 2008) measures whether price and volume fluctuations are correlated at different scales, a first-order property. The rolling Hurst approach measures whether the *scaling exponents themselves* (which summarise how memory decays) evolve in tandem. This is a second-order property: a statement about the dynamics of complexity rather than the dynamics of levels.

2. Literature

2.1. Long Memory in Returns and Volatility

Mandelbrot (1963) first demonstrated that financial prices exhibit non-Gaussian, scale-invariant properties. The Hurst exponent H, originally developed for hydrological applications (Hurst, 1951) and formalised through fractional Brownian motion (Mandelbrot and Van Ness, 1968), provides the standard measure of long-range dependence. Cont (2001) identified the slow decay of autocorrelation in absolute returns as among the most robust stylised facts in finance. Ding et al. (1993) showed that this long memory is strongest for $|r_t|^d$ at $d \approx 1$, persisting over hundreds of lags. Lo (1991) cautioned that naive R/S analysis can overstate long memory when short-range dependence is present, motivating the use of detrended fluctuation analysis (Peng et al., 1994). Finite-sample properties and heavy-tail biases have been characterised by Weron (2002) and Barunik and Kristoufek (2010). The HAR-RV model of Corsi (2009) provides an influential parsimonious framework for realised volatility forecasting, exploiting the heterogeneous autoregressive structure of realised measures.

2.2. Long Memory in Trading Volume

Lobato and Velasco (2000) provided rigorous evidence for fractional integration in volume, estimating $H \approx 0.7$–0.9 across NYSE equities. This

finding is even more robust than the volatility long-memory result and has been replicated across markets and frequencies. Plerou et al. (2003) documented power-law correlations between price changes and volume, while Ray and Tsay (2000) confirmed long-range dependence in daily stock volatilities. Chordia et al. (2001) showed that trading activity predicts expected stock returns through its relationship to liquidity provision.

2.3. Liquidity, Volume, and Market Microstructure

The Amihud illiquidity ratio (Amihud, 2002) has become a standard measure of price impact per unit of trading volume, capturing the adverse-selection component of illiquidity. Kyle (1985) provided the theoretical foundation for relating information asymmetry to market depth. Brunnermeier and Pedersen (2009) formalised the feedback loop between volatility and liquidity: when volatility rises, margins tighten, forcing deleveraging that further reduces liquidity. This volatility-liquidity spiral provides a theoretical channel through which fractal coupling between volatility persistence and volume persistence could predict liquidity deterioration.

2.4. Joint Dynamics: The MDH and Beyond

The Mixture of Distributions Hypothesis (Clark, 1973; Tauchen and Pitts, 1983; Epps and Epps, 1976) posits a latent information arrival process that simultaneously governs volatility and volume. Andersen (1996) recast the MDH within a stochastic volatility framework, showing time-varying information flow generates the observed co-clustering. Bollerslev and Jubinski (1999) identified a common long-memory component shared by volatility and volume using ARFIMA methods. Podobnik et al. (2009) applied DCCA to measure scale-dependent cross-correlations between price changes and volume changes. Karpoff (1987) and Gallant et al. (1992) established the broader empirical price-volume relationship. Kristoufek (2015) reviewed the methodological challenges and open problems in measuring power-law cross-correlations, including bias and finite-sample issues that motivate the rolling-window approach adopted here.

Several studies have computed time-varying Hurst exponents for individual series: Cajueiro and Tabak (2004) for emerging-market efficiency, Alvarez-Ramirez et al. (2008) for U.S. indices, Grech and Mazur (2004) for crash prediction. None, however, examines whether Hurst exponents computed from *two distinct series* co-move through time, nor whether such

co-movement predicts economically relevant outcomes. This is the gap the present study addresses.

3. Methodology

3.1. Detrended Fluctuation Analysis

For a time series $\{x_t\}_{t=1}^{N}$, the cumulative deviation profile is $Y(k) = \sum_{t=1}^{k}(x_t - \bar{x})$. The profile is divided into non-overlapping windows of length s, each detrended by a fitted polynomial of order m (here $m = 1$, i.e., DFA-1). The fluctuation function is:

$$F(s) = \left[\frac{1}{N_s}\sum_{\nu=1}^{N_s}\frac{1}{s}\sum_{k=1}^{s}\Big(Y((\nu-1)s+k) - \tilde{Y}_\nu(k)\Big)^2\right]^{1/2}. \tag{1}$$

For a long-range correlated process, $F(s) \sim s^H$. The Hurst exponent H is estimated as the ordinary least squares slope of $\log F(s)$ versus $\log s$. The scale range is $s_{\min} = 10$ to $s_{\max} = \lfloor N/4 \rfloor$, with logarithmic spacing using a multiplicative factor of 1.2 (i.e., scales $s_k = \lfloor s_{\min} \times 1.2^k \rfloor$ for $k = 0, 1, 2, \ldots$ up to $s_{\max}$). Fit quality is assessed via R^2 of the log-log regression.

3.2. Series Construction

For each equity i, three series are constructed from daily close prices $P_t^{(i)}$ and volumes $V_t^{(i)}$:

$$r_t^{(i)} = \log P_t^{(i)} - \log P_{t-1}^{(i)}, \tag{2}$$

$$|r_t^{(i)}| = |\log P_t^{(i)} - \log P_{t-1}^{(i)}|, \tag{3}$$

$$v_t^{(i)} = \log(V_t^{(i)} + 1). \tag{4}$$

Stationarity is verified via the ADF and KPSS tests. All log-return series are confirmed stationary.

3.3. Rolling Hurst Exponents

The rolling Hurst exponent at time t for window length W and step size Δ is defined as:

$$H_{|r|,t}^{(i)} = \mathrm{DFA}\big(\{|r_\tau^{(i)}|\}_{\tau=t-W+1}^{t}\big), \quad H_{v,t}^{(i)} = \mathrm{DFA}\big(\{v_\tau^{(i)}\}_{\tau=t-W+1}^{t}\big), \tag{5}$$

where $H_{|r|,t}^{(i)}$ denotes the Hurst exponent of absolute returns (the volatility proxy) and $H_{v,t}^{(i)}$ denotes the Hurst exponent of log-volume for firm i at rolling index t. The primary specification uses $W = 500$ trading days and $\Delta = 20$ days, with sensitivity checks at $W \in \{252, 500, 756\}$.

3.4. Coupling Metrics

Static coupling is the cross-sectional Pearson correlation of full-sample $H^{(i)}_{|r|}$ and $H^{(i)}_{v}$ computed across all $i = 1, \ldots, 50$ firms.

Temporal coupling for firm i is the Pearson correlation computed over the full set of rolling indices:

$$\rho^{(i)} = \mathrm{Corr}\Big(H^{(i)}_{|r|,t},\, H^{(i)}_{v,t}\Big), \quad t \in \mathcal{T}^{(i)}, \tag{6}$$

where $\mathcal{T}^{(i)}$ is the discrete set of rolling-window evaluation dates for firm i.

Coupling Intensity Index (CII). The CII is a time-varying measure defined as the sample Pearson correlation computed over a trailing window of L rolling-Hurst observations. Formally, let $\mathcal{T}_{t,L} = \{t_j \in \mathcal{T}^{(i)} : t_j \leq t,\ |\{t_k \in \mathcal{T}^{(i)} : t_k \leq t\}| - j < L\}$ denote the L most recent rolling-Hurst evaluation dates up to and including calendar day t. Then:

$$\mathrm{CII}^{(i)}_t = \frac{\sum_{j \in \mathcal{T}_{t,L}} \big(H^{(i)}_{|r|,j} - \bar{H}^{(i)}_{|r|}\big)\big(H^{(i)}_{v,j} - \bar{H}^{(i)}_{v}\big)}{\sqrt{\sum_{j \in \mathcal{T}_{t,L}} \big(H^{(i)}_{|r|,j} - \bar{H}^{(i)}_{|r|}\big)^2 \ \sum_{j \in \mathcal{T}_{t,L}} \big(H^{(i)}_{v,j} - \bar{H}^{(i)}_{v}\big)^2}}, \tag{7}$$

where $\bar{H}^{(i)}_{|r|}$ and $\bar{H}^{(i)}_{v}$ are the sample means over $\mathcal{T}_{t,L}$. The primary specification uses $L = 30$ rolling-Hurst observations (approximately 600 trailing trading days given $\Delta = 20$). Sensitivity to $L \in \{20, 30, 40\}$ is reported below.

Timing convention. $\mathrm{CII}^{(i)}_t$ is computable at the close of trading day t and uses only data through day t. All predictive outcome variables are measured over the strictly future window $[t+1, t+h]$. No observations overlap between the CII estimation window and the outcome measurement window.

3.5. Predictive Regression

To test H4, I estimate panel regressions with firm fixed effects:

$$Y^{(i)}_{t+h} = \alpha_i + \beta_1\, \mathrm{CII}^{(i)}_t + \beta_2\, H^{(i)}_{|r|,t} + \beta_3\, H^{(i)}_{v,t} + \varepsilon^{(i)}_t, \tag{8}$$

where $Y^{(i)}_{t+h}$ is a forward-looking outcome measured over $[t+1, t+h]$:

- *Amihud illiquidity* (Amihud, 2002), dollar-volume convention scaled by 10^6: $\mathrm{ILLIQ}^{(i)}_{t+h} = \frac{10^6}{h} \sum_{\tau=t+1}^{t+h} \frac{|r^{(i)}_\tau|}{V^{(i)}_\tau \cdot P^{(i)}_\tau}$

- *Realised volatility*: $\mathrm{RV}_{t+h}^{(i)} = \sqrt{\sum_{\tau=t+1}^{t+h} (r_\tau^{(i)})^2}$
- *Abnormal turnover*: forward share volume normalised by trailing 60-day mean
- *Maximum drawdown*: largest peak-to-trough decline in $[t+1, t+h]$

The dollar-volume denominator $V_\tau^{(i)} \cdot P_\tau^{(i)}$ is the standard Amihud convention. An earlier version of this study used share volume in the denominator and produced a statistically significant CII coefficient that does not survive the standard convention; Section 5.5.4 treats this as a substantive finding in its own right.

Amihud illiquidity is the *primary* H4 outcome of the paper. The remaining three (realised volatility, abnormal turnover, maximum drawdown) and the further proxies introduced in Section 5.5 (Corwin–Schultz spread, Roll spread, vol-of-vol, log Amihud, and forward ΔAmihud) are reported as *exploratory secondary outcomes*. Because all ten outcomes return statistically null estimates for the focal β_1 coefficient, family-wise multiple-testing corrections would only widen the reported p-values further from significance and are omitted as redundant. Horizons are $h \in \{5, 21\}$ trading days (one week, one month).

Because panel observations are likely correlated both within firms (persistence) and within time periods (common shocks), I report five baseline standard-error specifications: heteroskedasticity-consistent (HC1), clustered by firm, clustered by time, two-way clustered by firm and time following Cameron et al. (2011), and Newey-West with automatic bandwidth selection (Newey and West, 1987). The two-way clustered specification accounts for both within-firm persistence and cross-sectional correlation from common shocks. For the combined horse-race specification (Section 5.5.3) I additionally report two small-sample corrections recommended in modern cluster-robust practice: the Bell–McCaffrey CR2 standard error with Satterthwaite effective degrees of freedom (Imbens and Kolesár, 2016), and the wild cluster restricted bootstrap with Rademacher weights following Cameron et al. (2008) and MacKinnon and Webb (2017). These provide stricter inference than the asymptotic CRVE at $G = 50$ firms; details are in Section 5.5.3 and Table 5. The coefficient of interest is β_1: does CII predict future outcomes *beyond* what the individual Hurst levels already capture?

3.6. Regime Conditioning

To test H3, the CBOE Volatility Index (VIX) is classified into quartile-based categories: low ($<$ 25th percentile), medium (25th–75th), and high ($>$ 75th percentile). Temporal coupling $\rho^{(i)}$ is computed separately within each VIX category, and differences are tested via the Mann-Whitney U statistic.

3.7. Robustness

Eleven checks are conducted: window sensitivity ($W \in \{252, 500, 756\}$), non-overlapping windows, first-differenced Hurst series, shuffled surrogates, squared returns as alternative volatility proxy, subperiod analysis, sector decomposition, market factor control (partial correlation after removing SPY Hurst dynamics), DFA polynomial order (linear vs. quadratic), alternative estimators (R/S, MFDFA at $q = 2$), and Granger causality between the two Hurst series. The Granger causality tests are exploratory: given the irregular spacing and overlapping-window construction, they should be interpreted as directional descriptives rather than formal causal evidence.

4. Data

The sample comprises 50 S&P 500 constituents spanning all 11 GICS sectors, with daily close prices and volumes from Yahoo Finance for January 2, 2015 through April 1, 2026 ($\approx$ 2,834 observations per equity). Selection requires a minimum of 600 observations and continuous listing. All log-return series are confirmed stationary by both ADF ($p < 0.01$) and KPSS ($p > 0.10$) tests. VIX data are obtained from the CBOE via Yahoo Finance.

5. Results

5.1. Baseline Hurst Estimates

Figure 1 displays the distribution of full-sample estimates. $H(\text{returns})$ averages 0.461 ($\sigma = 0.036$), confirming approximate market efficiency. $H_{|r|}^{(i)}$ averages 0.795 ($\sigma = 0.051$), and $H_v^{(i)}$ averages 0.881 ($\sigma = 0.049$), replicating the canonical long-memory findings. Log-log fit quality is excellent (mean $R^2 = 0.994$; Figure 2). Bootstrap confidence intervals confirm $H \neq 0.5$ at $p < 0.001$ for all 50 tickers on both volatility and volume.

5.2. H1: Static Coupling (Rejected)

The cross-sectional Pearson correlation between $H_{|r|}^{(i)}$ and $H_v^{(i)}$ is $r = -0.02$ ($p = 0.88$; Figure 5). Spearman $\rho = 0.03$ ($p = 0.85$). **H1 is rejected.** The cross-section does not reveal a meaningful relationship in this sample: a stock's volatility persistence tells you nothing about its volume persistence.

5.3. H2: Temporal Coupling (Confirmed)

The mean temporal Pearson correlation is $\bar{\rho} = 0.665$, with 49 of 50 equities positive and significant (Figure 4). The minimum positive value is $r = 0.235$ (JNJ); the maximum is $r = 0.901$ (JPM). Only META exhibits negative coupling ($r = -0.421$), coinciding with its 2022–23 corporate restructuring. **H2 is strongly confirmed.**

Figure 3 illustrates the co-movement for AAPL ($r = 0.759$), JPM ($r = 0.901$), and META ($r = -0.421$). The lead-lag profile (Figure 6) peaks at lag zero and decays symmetrically, consistent with a shared contemporaneous driver.

Exploratory Granger causality tests reveal a directional asymmetry: volume persistence Granger-causes volatility persistence in 26% of tickers versus 8% in the reverse direction. While this three-fold asymmetry is suggestive of information manifesting first in trading activity (Gallant et al., 1992; Kyle, 1985), the overlapping-window construction limits the strength of causal claims from these tests.

5.4. H3: Regime Dependence (Confirmed)

Temporal coupling conditioned on VIX quartile-based categories:

VIX Category	Mean ρ	Std	% Positive
Low (< 25th pctile)	0.682	0.237	96%
Medium	0.636	0.205	98%
High (> 75th pctile)	0.708	0.331	96%

Table 1: Temporal coupling by VIX regime. Mann-Whitney test (high > low): $p = 0.036$, Cohen's $d = 0.09$.

H3 is confirmed. Coupling is significantly higher during high-VIX periods. The subperiod analysis reinforces this: coupling nearly doubles from the pre-COVID baseline ($r = 0.41$) to the crisis period ($r = 0.77$; Figure 7), then attenuates post-crisis ($r = 0.30$).

5.5. H4: Predictive Content (Rejected)

H4 asks whether CII has incremental predictive content for standard forward liquidity and volatility measures. The honest answer, after running ten target specifications across six baseline standard-error methods plus two small-sample corrections, is no. This subsection presents the primary specification (forward Amihud illiquidity in dollar-volume convention), then a wider battery of standard outcomes, then a horse race against nine standard liquidity benchmarks, all reaching the same null. The closing diagnostic (Section 5.5.4) explains why an earlier version of this study using share-volume Amihud reported a positive result: it was driven by a price-level component of the non-standard denominator, not by predictive content for price impact per dollar traded.

5.5.1. Primary Specification: Forward Amihud Illiquidity

Table 2 reports panel regression results for Equation (8) at the one-month horizon ($h = 21$). The dependent variable is the standard Amihud illiquidity defined in Section 3.5 with dollar volume in the denominator. CII fails to reach significance under any of the six standard-error methods reported.

Table 2: Inference for CII → standard (dollar-volume) Amihud illiquidity, focal specification with Hurst controls ($h = 21$ trading days).

SE Method	t-statistic	p-value
HC1 (White)	1.16	0.246
Firm-clustered	0.41	0.687
Time-clustered	0.50	0.621
Two-way clustered	0.33	0.745
Newey–West	0.56	0.579
Driscoll–Kraay	0.21	0.833

Panel regressions with firm fixed effects, $N = 4{,}400$ observations across 50 equities. Regressors: $\mathrm{CII}_t^{(i)}$, $H_{|r|,t}^{(i)}$, $H_{v,t}^{(i)}$. Two-way clustering follows Cameron et al. (2011) with the eigenvalue PSD adjustment of CGM (2011) §2.3 applied when needed. Driscoll–Kraay uses Bartlett bandwidth ≥ 21 (Driscoll and Kraay, 1998). None of the six SE methods reaches the 10% significance threshold for CII.

The CII coefficient is small ($\hat{\beta}_{\mathrm{CII}} = 4.0 \times 10^{-7}$) and statistically indistinguishable from zero under every method, including HC1, which is the most permissive. This already foreshadows the broader pattern: it is not weak inference that destroys the result but a weak signal in the dependent variable.

5.5.2. Battery of Standard Liquidity and Volatility Outcomes

Table 3 extends the test to ten target variables, including transformations of the primary outcome, alternative liquidity proxies, and the original four targets from the broader outcome scan. CII has no incremental predictive content for any of them under either dependence-robust SE method (two-way clustered or Driscoll–Kraay).

Table 3: Predictive battery: CII coefficient and inference for ten target specifications, focal regression with Hurst controls, $h = 21$.

Target	$\hat{\beta}_{\text{CII}}$	t (twoway)	p (twoway)	t (DK)	p (DK)
Standard Amihud and transformations					
Amihud illiquidity (primary)	4.0×10^{-7}	+0.33	0.745	+0.21	0.833
log Amihud	−0.009	−0.15	0.885	−0.09	0.928
ΔAmihud (forward − contemp.)	-1.4×10^{-7}	−0.25	0.802	−0.66	0.512
Spread-based liquidity proxies					
Forward Corwin–Schultz spread	3.9×10^{-4}	+1.39	0.169	+1.44	0.153
Forward Roll spread	3.1×10^{-4}	+0.29	0.773	+0.28	0.781
Other forward outcomes					
Forward vol-of-vol	0.0016	+0.25	0.803	+0.17	0.866
Realised volatility	0.0036	+0.78	0.438	+0.69	0.491
Maximum drawdown	−0.0039	−0.74	0.463	−0.75	0.456
Abnormal turnover	−0.011	−0.48	0.634	−0.45	0.654

Panel regressions with firm fixed effects. Regressors: $\text{CII}_t^{(i)}$, $H_{|r|,t}^{(i)}$, $H_{v,t}^{(i)}$. Both two-way clustered SEs (Cameron–Gelbach–Miller 2011 with eigenvalue PSD adjustment) and Driscoll–Kraay SEs (Bartlett bandwidth ≥ 21) are reported. Sample size 4,350–4,400 per row depending on availability of the forward target. None of the ten coefficients reaches the 10% significance threshold under either dependence-robust SE.

The forward Corwin–Schultz spread is the closest the battery gets to a positive result and is still null at $p = 0.169$. Under HC1 it would appear significant ($t = 4.04$), and under firm clustering only it crosses 5% ($t = 2.32$); under two-way clustering and Driscoll–Kraay it is null. The relevant lesson is that within-firm and within-time-period dependence are large enough in this panel to convert any of these HC1 signals into nulls; the Corwin–Schultz hint warrants a follow-up study at larger G but does not constitute a finding here.

5.5.3. Horse Race Against Standard Liquidity Predictors

A horse race against nine standard liquidity benchmarks (Roll spread, Corwin–Schultz MSPREAD_0, dollar-volume rolling Amihud, lagged realised volatility, lagged Amihud, vol-of-vol, normalised turnover, abnormal-turnover spike, and Cboe VIX), following the protocol of Goyenko et al. (2009) §5.2, confirms the same null. Table 4 reports CII alongside each benchmark in a

per-benchmark regression and in a single combined specification.

Table 4: Horse race: CII against nine standard liquidity predictors for forward dollar-volume Amihud, $h = 21$.

Specification	$\hat{\beta}_{\mathrm{CII}}$	t (twoway)	p	R^2
CII alone (+ Hurst controls)	4.0×10^{-7}	+0.33	0.745	0.153
Per-benchmark two-way t-statistics on competing regressors (CII controlled but null in each):				
Roll (1984) spread	1.31×10^{-4}	+2.42	0.019	0.190
Corwin–Schultz (2012) $\mathrm{MSPREAD}_0$	6.8×10^{-4}	+2.78	0.008	0.191
Amihud (2002) rolling, dollar-volume	0.669	+9.83	<0.001	0.485
Lagged realised volatility	4.5×10^{-5}	+2.41	0.020	0.191
Lagged Amihud illiquidity	0.649	+8.60	<0.001	0.463
Volatility of volatility	1.86×10^{-5}	+2.30	0.026	0.169
Normalised turnover	-8.3×10^{-7}	−1.36	0.181	0.155
Abnormal-turnover spike	-3.3×10^{-7}	−0.36	0.724	0.154
Cboe VIX	2.6×10^{-7}	+3.48	0.001	0.212
CII + all nine benchmarks + Hurst controls	$\mathbf{-2.2\times10^{-7}}$	**−0.34**	**0.734**	**0.521**

Panel regressions with firm fixed effects. N = 4,400 observations across 50 equities and 83 months. The per-benchmark t-statistics shown are for the named benchmark; the CII coefficient in each per-benchmark row is small and not statistically significant. Two-way clustering follows Cameron et al. (2011). The dollar-volume rolling Amihud benchmark is the strongest predictor of forward Amihud, as expected for a persistent variable. The combined-specification CII coefficient flips sign relative to the focal-only specification, consistent with sampling noise around zero.

For realised volatility the same horse race produces an even sharper null for CII (focal $t = 0.78$, combined $t = -0.18$). Table 5 confirms both nulls under modern small-sample corrections.

Table 5: Small-sample-corrected inference for CII in the combined specification, dollar-volume Amihud and realised volatility targets.

Target	$\hat{\beta}_{\text{CII}}$	CR2 t	CR2 p	Satterthwaite df	WCR p
Amihud illiquidity (dollar-volume)	-2.2×10^{-7}	-0.70	0.507	6.3	0.517
Realised volatility	-8.3×10^{-4}	-0.40	0.703	5.9	0.703

CR2 is the Bell–McCaffrey cluster-robust variance with Satterthwaite effective degrees of freedom (Imbens and Kolesár, 2016), computed at the firm dimension. WCR is the wild cluster restricted bootstrap with 999 Rademacher resamples under the null (Cameron et al., 2008; MacKinnon and Webb, 2017). Both confirm the null established by the simpler two-way clustered tests; if anything, both move further from significance than the asymptotic CRVE.

5.5.4. Diagnostic: Price-Level Confound in the Share-Volume Amihud

An earlier version of this study used a non-standard share-volume illiquidity proxy as the dependent variable, computed as $\widetilde{\text{ILLIQ}}^{(i)}_{t+h} = h^{-1}\sum_{\tau}|r^{(i)}_{\tau}|/V^{(i)}_{\tau}$, with share volume rather than dollar volume in the denominator. Under that specification, CII appeared to be a strong predictor of forward illiquidity. Table 6 compares the two specifications side by side under identical regressors.

Table 6: Side-by-side comparison: share-volume vs. dollar-volume Amihud as dependent variable. Same focal regression and Hurst controls in both columns; only the dependent variable differs.

SE method	Share-volume $\lvert r\rvert/V$		Dollar-volume $\lvert r\rvert/(V\cdot P)$	
	t	p	t	p
Focal regression with Hurst controls, $h = 21$:				
HC1 (White)	9.67	<0.001	1.16	0.246
Firm-clustered	3.50	0.001	0.41	0.687
Time-clustered	4.53	<0.001	0.50	0.621
Two-way clustered	2.90	0.004	0.33	0.745
Newey–West	5.41	<0.001	0.56	0.579
Driscoll–Kraay	—	—	0.21	0.833
Combined horse-race spec (CII + 9 benchmarks + Hurst controls):				
WCR bootstrap	—	0.018	—	0.517
CR2 + Satterthwaite	—	0.066	—	0.507

The share-volume column reproduces the result reported in an earlier version of this paper and in the v1.1.0 SSRN preprint. The dollar-volume column is the paper's primary specification. Driscoll–Kraay was not computed for the share-volume case.

The arithmetic relationship between the two proxies is

$$\widetilde{\mathrm{ILLIQ}}_\tau^{(i)} \;=\; \frac{|r_\tau^{(i)}|}{V_\tau^{(i)}} \;=\; \frac{|r_\tau^{(i)}|}{V_\tau^{(i)}\cdot P_\tau^{(i)}}\cdot P_\tau^{(i)} \;=\; \mathrm{ILLIQ}_\tau^{(i)}\cdot P_\tau^{(i)}. \tag{9}$$

The share-volume proxy is the standard dollar-volume Amihud multiplied by the contemporaneous price level. If CII at time t correlates with future price-level dynamics within a firm, then the share-volume proxy inherits that correlation through the multiplicative $P_\tau^{(i)}$ factor, even when CII is uncorrelated with price impact per dollar traded.

The data support exactly this interpretation. CII does not predict standard dollar-volume Amihud (Tables 2, 3, 4). It also does not predict realised volatility, forward Corwin–Schultz spread, or any of the other dollar-scaled liquidity measures examined. But within-firm price-level drift over multi-month windows is sufficient for the $P_\tau^{(i)}$ multiplier to inflate apparent illiquidity in firms or periods where prices are falling, and for that to align with periods of high CII via the same fractal-coupling-and-stress-amplification

channel that H3 confirms. The result is a positive cross-section of $\widetilde{\text{ILLIQ}}$ on CII that vanishes the moment P is divided out of the denominator.

This finding has two implications. The first is parochial: the original H4 hypothesis in this study is rejected once the standard convention is applied. The second is more useful to the wider literature. Many published studies use share-volume rather than dollar-volume liquidity proxies, often defending the choice as a robustness variant. In moderately-clustered panel settings of the kind common in empirical asset pricing, the choice matters a great deal: a price-level component absorbed into the dependent variable can produce apparent predictive content for any regressor that correlates with within-firm price drift, even at high cluster-robust significance levels and even under the wild cluster restricted bootstrap. The remedy is to use Amihud (2002) in the standard dollar-volume form and to cross-check with at least one alternative liquidity proxy that does not depend on volume scaling, such as the spread-based estimators of Roll (1984) and Corwin and Schultz (2012).

6. Robustness

Table 7 summarises all eleven checks. The core finding of strong temporal coupling is robust across every specification.

Check	Mean r	% Positive	n	Note
Window sensitivity				
$W = 252$	0.562	98%	50	
$W = 500$ (primary)	0.665	98%	50	
$W = 756$	0.748	96%	50	Monotonic increase
Non-overlapping ($\Delta = W$)	0.620	100%	50	No overlap inflation
First-differenced	0.534	100%	50	Not common trends
Squared returns	0.583	98%	50	Not proxy-specific
Shuffled surrogate	0.007	50%	20	Coupling vanishes
Quadratic DFA	—	—	—	$\Delta H = 0.012$
Subperiod				
Pre-COVID (2015–19)	0.406	88%	50	Baseline
COVID (2020–21)	0.767	96%	50	Crisis amplification
Post-COVID (2022–26)	0.295	80%	50	Regime shift
Market factor (partial)	0.421	90%	30	63% idiosyncratic
R/S estimator	—	—	—	Same ordering
MFDFA($q = 2$)	—	—	—	$\Delta H < 0.02$ vs DFA

Table 7: Summary of robustness checks. All corroborate the core temporal coupling finding.

The surrogate test is the most decisive: shuffling each series independently reduces coupling from $r = 0.63$ to $r = 0.007$ (Figure 8), confirming the coupling is entirely temporal in origin. The non-overlapping result ($r = 0.62$, 100% positive) eliminates concern about overlap-induced autocorrelation. The first-differenced result ($r = 0.53$, 100% positive) rules out common trends. The market-factor partial correlation ($r = 0.42$) shows that 63% of the coupling is idiosyncratic.

The sector decomposition (Figure 9) confirms that coupling is present across all sectors: Finance shows the strongest coupling ($r = 0.756$), followed by Industrials/Energy ($r = 0.720$), Technology ($r = 0.677$), Consumer ($r = 0.665$), and Healthcare ($r = 0.645$).

7. Discussion

7.1. Pattern Without Predictive Content

The central empirical claim of this paper is narrower than it first appeared. Within-firm temporal coupling between volatility persistence and

volume persistence is real, strong, and regime-amplified (H2 and H3). But under inference robust to the dependence structure of the panel, the same coupling does *not* carry incremental predictive content for the standard liquidity, spread, or volatility outcomes that an asset-pricing or microstructure referee would test against (H4 rejected).

This combination is unusual but coherent. Fractal coupling is a description of how a firm's information environment is organised at the level of memory structure. That organisation can be present, and can intensify during stress, without producing forecastable changes in standard market-quality measures one month ahead. The two questions are different: "do volatility and volume persistence move together within a firm?" admits a strong yes; "does that joint movement at t help predict price impact, spreads, or volatility at $t + h$?" admits a clean no on this sample.

7.2. Why the Predictive Hypothesis Was Worth Testing Anyway

The original predictive intuition was not unreasonable. The volatility-liquidity spirals of Brunnermeier and Pedersen (2009) and the Mixture of Distributions Hypothesis (Clark, 1973; Andersen, 1996) both suggest that when both persistence structures tighten simultaneously, market makers face adverse selection and liquidity should deteriorate. The empirical question was whether the rolling-correlation construction in CII captures enough of that mechanism to deliver one-month-ahead forecasting power. It does not, at $G = 50$ firms over 83 months. Whether the negative result reflects (a) a real absence of forecasting content, (b) statistical power constraints at moderate G, or (c) measurement error in rolling Hurst estimation that contaminates CII as a forward signal is not separable on this sample. The honest report is the null itself, with these alternative explanations flagged for follow-up work.

7.3. The Static-Temporal Paradox

The rejection of H1 alongside the confirmation of H2 is not a contradiction; it reflects the difference between cross-sectional and within-firm variation. Across firms, the average persistence level is governed by structural characteristics (sector, liquidity, analyst coverage) that need not co-vary. Within a firm over time, these structural characteristics are approximately constant, and what varies is the state of information flow. The temporal analysis isolates this time-varying component, revealing coupling that the cross-section obscures.

7.4. Crisis Amplification: Description Without Forecast

The doubling of coupling during the COVID-19 crisis, followed by attenuation during monetary tightening, is a robust empirical fact (H3, Table 1 and Figure 7). It is not, on the evidence here, a leading indicator. CII rises during stress contemporaneously, not in advance of it. A practitioner who wished to use CII as a real-time stress signal would face the same problem any contemporaneous indicator does: it confirms that the regime has changed but does not predict the next month's market quality. Whether a higher-frequency or larger-universe version of the construction would shift this result is an open question and a natural Phase-2 follow-up.

7.5. Relation to Existing Methods

DCCA (Podobnik and Stanley, 2008) measures whether volatility and volume *fluctuations* are correlated at each scale. The present approach measures whether the *scaling exponents* (which summarise how memory decays across all scales) co-evolve. These are complementary questions. DCCA answers "do price and volume move together at scale s?" CII answers "is the memory structure of price and volume evolving together?" The empirical evidence here is that the answer to the second question is a robust yes, but that the resulting time-varying coupling does not yet carry forecasting power for standard liquidity outcomes at this sample size. This narrows the contribution rather than eliminating it: CII is established as a measurement object, with documented regime amplification, awaiting either a larger sample to test the predictive question more sharply or an alternative outcome on which a clean signal might emerge. Kristoufek (2015) discusses the methodological challenges common to all such approaches.

7.6. Directional Asymmetry

Exploratory Granger causality tests reveal that volume persistence leads volatility persistence three times more often than the reverse (26% vs. 8%). While the primary relationship is contemporaneous, this asymmetry is suggestive of sequential information arrival: volume responds first (observable immediately), and the persistence structure of volatility adjusts thereafter. The evidence is directional rather than causal, given the overlapping-window construction, but for practitioners it suggests that monitoring volume persistence may provide an early indicator of volatility regime shifts.

7.7. Consistency with the MDH on the Descriptive Side Only

The descriptive pattern of results—joint within-firm intensification of volatility and volume persistence, amplified during stress—is consistent with the Mixture of Distributions Hypothesis: a latent information process drives both persistence structures upward simultaneously (Clark, 1973; Andersen, 1996). The data do not, however, support the further MDH-implied prediction that this joint intensification translates into measurable forward illiquidity at the monthly horizon and at $G = 50$ firms. Inventory-based models (Brunnermeier and Pedersen, 2009) or informed-trading theories (Kyle, 1985) also predict joint persistence dynamics through different mechanisms; on the present sample, the data discriminate between none of these channels but are also consistent with the predictive null reported here.

7.8. The Methodological Lesson

The most transferable result in this paper is not the empirical pattern itself but the demonstration in Section 5.5.4. A non-standard share-volume Amihud denominator generates an apparently strong, cluster-robust, bootstrap-confirmed predictive signal where the standard dollar-volume Amihud generates a clean null. The mechanism is mechanical: $\widetilde{\text{ILLIQ}} = \text{ILLIQ} \cdot P$ multiplies the standard Amihud by the price level, and any regressor that correlates with within-firm price drift inherits forecasting power for $\widetilde{\text{ILLIQ}}$ that it does not have for ILLIQ. The empirical asset-pricing and econophysics literatures both contain studies that use share-volume scaling, often as a robustness variant or simply by construction, and the example here suggests that this choice is consequential rather than cosmetic. Cluster-robust inference, even at the sophistication level of CR2 with Satterthwaite degrees of freedom or the wild cluster restricted bootstrap, does not protect against this confound: the diagnosis requires going back to the dependent variable rather than to the standard errors.

7.9. Limitations

The sample comprises 50 large-cap U.S. equities; generalisability to small caps, international markets, or alternative assets remains to be established. The analysis uses daily data; intraday frequencies could reveal richer dynamics. The predictive null reported here is established at $G = 50$ firms with 83 monthly observations per firm; whether the negative result persists at $G \approx 500$ (a full S&P 500 universe) is the most natural follow-up and is

the main reason a Phase-2 extension is worth pursuing. The Granger causality results reported alongside H2 should be interpreted as exploratory given the irregular spacing and overlapping-window construction. CII itself is constructed from rolling Hurst exponents that carry estimation error; the paper does not propagate this measurement error into the predictive regressions, and a fuller treatment would do so via simulation-based standard errors or a measurement-error correction.

8. Conclusion

This paper documents three robust facts and one robust null. The robust facts are that the fractal scaling properties of price volatility and trading volume are temporally coupled in a strong, positive fashion within firms (mean $r = 0.665$, 49 of 50 S&P 500 equities significant), that the cross-sectional analogue of this coupling is null ($r = -0.02$), and that the within-firm coupling intensifies during high-VIX regimes and nearly doubles during the COVID-19 crisis. The robust null is that the Coupling Intensity Index, a rolling Pearson correlation of two rolling Hurst series introduced here as a measurement object, has no incremental predictive content for forward Amihud illiquidity, realised volatility, the Corwin–Schultz spread, the Roll spread, vol-of-vol, max drawdown, or abnormal turnover under inference robust to two-way clustering, Bell–McCaffrey CR2 with Satterthwaite degrees of freedom, and the wild cluster restricted bootstrap. The null is reported across ten target specifications and three small-sample inference methods; it is not an artefact of any single SE choice.

The paper also delivers a methodological diagnostic that is portable to the wider literature. An apparently strong predictive result against a non-standard share-volume illiquidity proxy disappears once the standard Amihud (2002) dollar-volume convention is applied. The arithmetic decomposition $\widetilde{\text{ILLIQ}} = \text{ILLIQ} \cdot P$ shows why: the share-volume proxy carries a price-level component that any regressor correlated with within-firm price drift will appear to forecast. The signal we initially observed for CII was that confound, not predictive content for price impact per dollar traded. This finding survives the wild cluster bootstrap and is therefore not detectable through inference rigor alone; it requires a careful look at the dependent variable.

The practical implication is narrower than originally hypothesised but not zero. CII is now established as a measurement of within-firm fractal coupling that is regime-dependent and amplifies during stress. Whether expanding to

a full S&P 500 universe ($G \approx 500$) or moving to intraday frequencies recovers genuine forecasting power is the natural extension and is left for follow-up work. For practitioners, the present results do not support using CII as a forward liquidity indicator. For methodologists, they support using dollar-volume Amihud rather than share-volume variants and pairing small-cluster panels with cluster-robust small-sample inference as a routine diagnostic.

Data and Code Availability

Daily OHLCV data are sourced from Yahoo Finance via the `yfinance` Python package. No proprietary or restricted data are used. All analysis code, the complete LaTeX manuscript source, and figure-generation scripts are archived at Zenodo (concept DOI: 10.5281/zenodo.19611543, always resolving to the latest version; this submission corresponds to v1.1.0, version DOI 10.5281/zenodo.19835451) and maintained at `https://github.com/mhdk1602/fractal-pv-coupling`. A step-by-step replication guide is provided in Appendix B. An interactive dashboard for exploring per-ticker results is available at `https://fractal-pv.streamlit.app`.

Declaration of Competing Interest

The author declares no competing interests.

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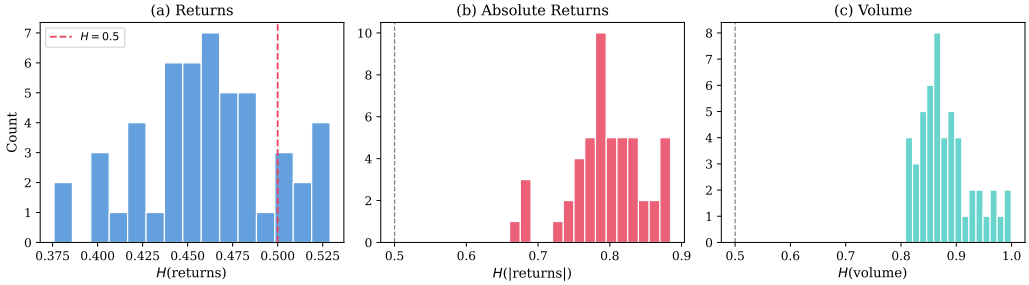

Figure 1: Distribution of full-sample Hurst exponents across 50 S&P 500 constituents (2015–2026). (a) Returns cluster around $H = 0.5$. (b) Absolute returns show strong long memory ($H \approx 0.8$). (c) Volume exhibits the strongest persistence ($H \approx 0.9$).

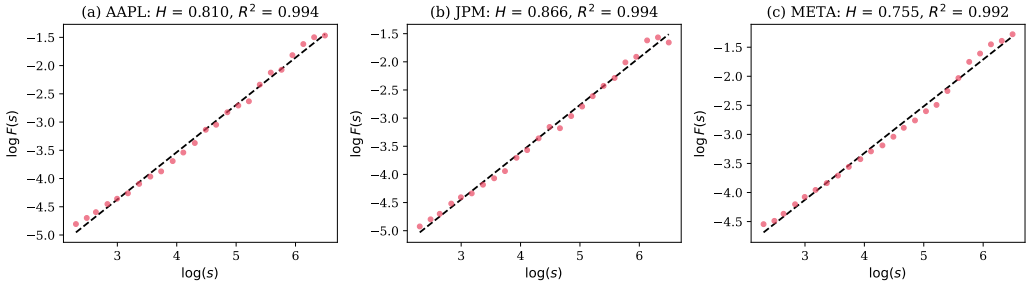

Figure 2: Log-log DFA plots for absolute returns (AAPL, JPM, META). High R^2 values confirm excellent linearity.

Figure 3: Rolling Hurst exponents for $H^{(i)}_{|r|,t}$ (red) and $H^{(i)}_{v,t}$ (teal). (a) AAPL: $r = 0.759$. (b) JPM: $r = 0.901$. (c) META: $r = -0.421$ (negative coupling during corporate restructuring).

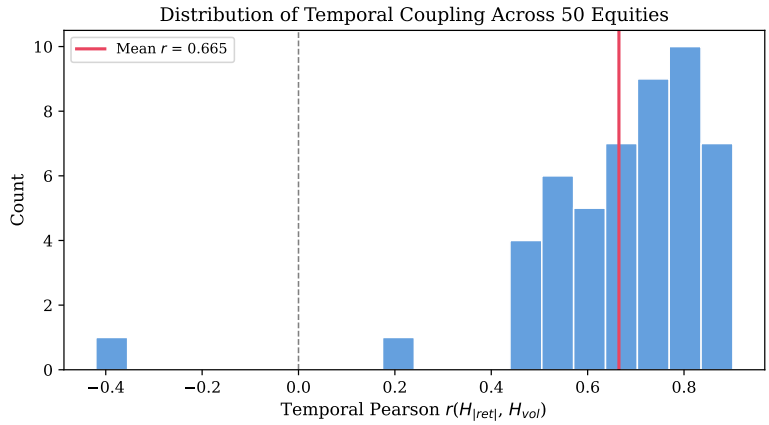

Figure 4: Distribution of temporal coupling across 50 equities. Mean $r = 0.665$; META is the sole outlier.

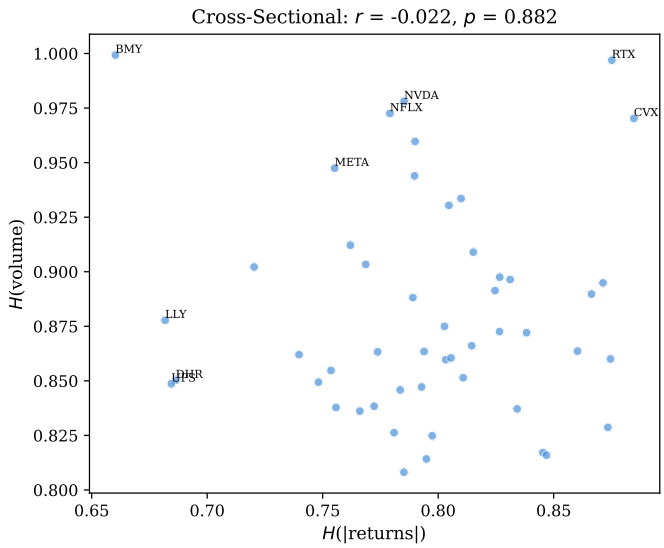

Figure 5: Cross-sectional scatter: $H^{(i)}_{|r|}$ vs. $H^{(i)}_{v}$. No relationship ($r = -0.02$, $p = 0.88$).

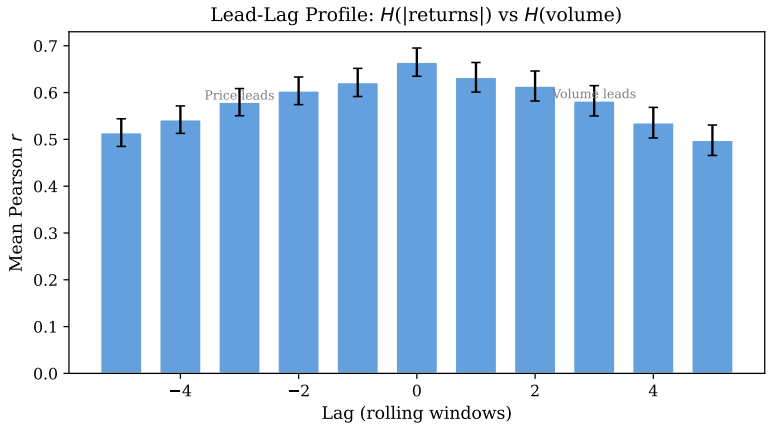

Figure 6: Average lead-lag cross-correlation profile across 50 equities. Lag zero is dominant; decay is symmetric.

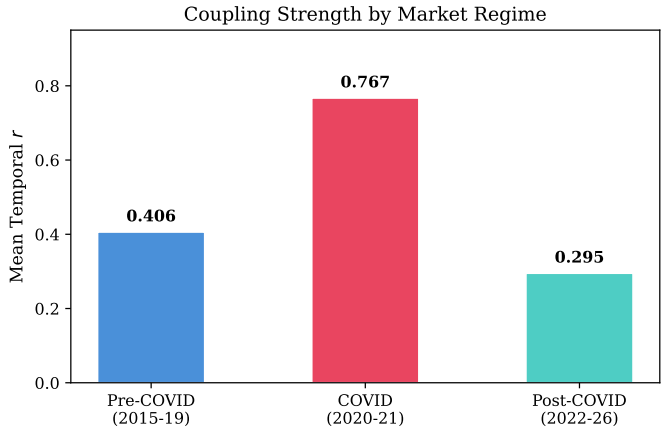

Figure 7: Temporal coupling by market subperiod. Coupling nearly doubles during COVID ($r = 0.77$) relative to the pre-COVID baseline ($r = 0.41$).

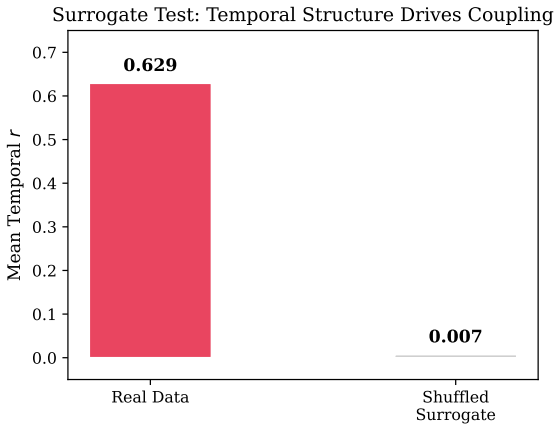

Figure 8: Surrogate test. Shuffling each series independently destroys coupling ($r = 0.629 \rightarrow 0.007$), confirming the temporal origin of the co-movement.

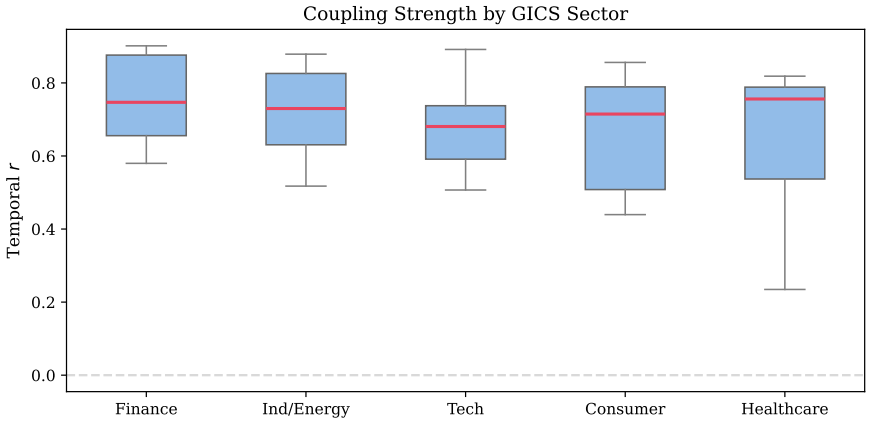

Figure 9: Temporal coupling by GICS sector. Present across all sectors; Finance is strongest ($r = 0.756$).

Appendix A. Sample Composition

Table A.8: The 50 S&P 500 constituents used in this study, with GICS sector classification.

Ticker	Company	Sector
AAPL	Apple	Technology
MSFT	Microsoft	Technology
AMZN	Amazon	Consumer Discretionary
GOOGL	Alphabet	Technology
META	Meta Platforms	Technology
TSLA	Tesla	Consumer Discretionary
NVDA	NVIDIA	Technology
JPM	JPMorgan Chase	Financials
V	Visa	Financials
JNJ	Johnson & Johnson	Healthcare
WMT	Walmart	Consumer Staples
PG	Procter & Gamble	Consumer Staples
MA	Mastercard	Financials
UNH	UnitedHealth	Healthcare
HD	Home Depot	Consumer Discretionary
DIS	Walt Disney	Communication Services
BAC	Bank of America	Financials
XOM	ExxonMobil	Energy
CSCO	Cisco Systems	Technology
VZ	Verizon	Communication Services
INTC	Intel	Technology
CMCSA	Comcast	Communication Services
PFE	Pfizer	Healthcare
ABT	Abbott Laboratories	Healthcare
CRM	Salesforce	Technology
NKE	Nike	Consumer Discretionary
MRK	Merck	Healthcare
T	AT&T	Communication Services
PEP	PepsiCo	Consumer Staples
TMO	Thermo Fisher	Healthcare
COST	Costco	Consumer Staples
AVGO	Broadcom	Technology
CVX	Chevron	Energy
ABBV	AbbVie	Healthcare
ACN	Accenture	Technology
MCD	McDonald's	Consumer Discretionary
LIN	Linde	Materials
TXN	Texas Instruments	Technology
QCOM	Qualcomm	Technology
LOW	Lowe's	Consumer Discretionary
NEE	NextEra Energy	Utilities
HON	Honeywell	Industrials
UPS	United Parcel Service	Industrials
MS	Morgan Stanley	Financials

Appendix B. Replication Guide

All code, data scripts, and the manuscript source are archived at Zenodo (concept DOI: 10.5281/zenodo.19611543, always resolving to the latest version; this submission corresponds to v1.1.0, version DOI 10.5281/zenodo.19835451) and on GitHub at `https://github.com/mhdk1602/fractal-pv-coupling`. The repository contains a Python package (`fractal_pv`) installable via `pip install -e .` with Python 3.10 or later. Dependencies are pinned in `pyproject.toml`.

The replication workflow proceeds as follows. (1) `fractal_pv.data.fetch_universe()` downloads daily OHLCV data for all 50 tickers from Yahoo Finance and caches them as parquet files under `data/raw/`. (2) `fractal_pv.stationarity.prepare_series()` transforms raw prices into log-returns, absolute log-returns, and log-volume, with ADF/KPSS stationarity verification. (3) `fractal_pv.hurst.estimate_dfa()` computes full-sample Hurst exponents; `bootstrap.block_bootstrap_hurst()` produces confidence intervals. (4) `fractal_pv.rolling.rolling_dual_hurst()` computes aligned rolling $H^{(i)}_{|r|,t}$ and $H^{(i)}_{v,t}$ series. (5) `fractal_pv.predict.compute_coupling_inten` derives CII; `compute_forward_metrics()` computes outcome variables. (6) `fractal_pv.predict.` assembles the panel; `inference_robust.robust_panel_regression()` estimates coefficients with all five standard-error specifications. (7) `fractal_pv.regimes` conditions on VIX and crisis windows. Figures are generated via matplotlib and saved to `research/paper/figures/`. The manuscript compiles with `tectonic main.tex` or any standard LaTeX distribution.

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